Mandatory deliverable

LAB | Setup Apache Server Lab

Skills you will learn

Version Control & Linux: Linux Fundamentals

1. Introduction

The **Apache HTTP Server** is one of the most widely used web servers in the world. It provides a powerful and flexible platform for hosting websites, featuring **dynamically loadable modules, strong media support, and seamless integration** with other software.

In this lab, you will learn how to **install and configure Apache on an Ubuntu Linux virtual machine** and ensure it is properly set up to serve web content.

2. Requirements

- **Ubuntu Linux Virtual Machine**
- **Administrative privileges** (sudo access)
- **Port 80 availability** (Ensure no other service like Nginx or another Apache instance is using port 80)

To stop conflicting services before proceeding, run:

```
sudo systemctl stop nginx
sudo systemctl stop apache2
```

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3. Installing Apache

3.1 Updating Package Lists

Before installing Apache, update the package lists to ensure you get the latest version:

```
sudo apt update -y
```

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Ask AI Tutor

3.2 Installing Apache

To install Apache, run:

```
sudo apt install apache2 -y
```

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This command will install **Apache and all necessary dependencies**.

3.3 Verifying Apache Installation

Once installed, Apache should start automatically. Check its status:

```
sudo systemctl status apache2
```

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If Apache is running, you should see output indicating that the service is **active**.

To test access, open a browser and visit:

```
http://<server_ip_address>
```

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If Apache is running correctly, you should see the **default Apache welcome page**.

4. Managing the Apache Process

Use the following `systemctl` commands to manage Apache:

- **Stop Apache:**

```
sudo systemctl stop apache2
```

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- **Start Apache:**

```
sudo systemctl start apache2
```

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- **Restart Apache:**

```
sudo systemctl restart apache2
```

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- **Reload Apache (apply configuration changes without restarting):**

```
sudo systemctl reload apache2
```

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- **Disable Apache from starting on boot:**

```
sudo systemctl disable apache2
```

- **Enable Apache to start on boot:**

```
sudo systemctl enable apache2
```

5. Setting Up Virtual Hosts

By default, Apache serves content from `/var/www/html`, which works for a **single website**. For multiple websites, use **Virtual Hosts**.

5.1 Creating a New Website Directory

Create a directory for your new website:

```
sudo mkdir -p /var/www/your_domain
```

Set the correct permissions:

```
sudo chmod -R 755 /var/www/your_domain
```

5.2 Creating an Index Page

Create an HTML file to serve as the homepage:

```
sudo nano /var/www/your_domain/index.html
```

Add the following content:

```
<html>
  <head>
    <title>Welcome to Your Domain!</title>
  </head>
  <body>
    <h1>Success! The your_domain virtual host is working!</h1>
  </body>
</html>
```

Save and exit the file.

5.3 Configuring a Virtual Host

Create a new Apache configuration file:

```
sudo nano /etc/apache2/sites-available/your_domain.conf
```

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Paste the following configuration:

```
<VirtualHost *:80>
    ServerAdmin webmaster@localhost
    ServerName your_domain
    ServerAlias www.your_domain
    DocumentRoot /var/www/your_domain
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

Copy Hide explanation

This block defines a separate Apache website using a Virtual Host, which is Apache's way of deciding, "when a request comes in for this name, serve files from this folder instead of the default site."

`<VirtualHost *:80>` means this site should answer requests on port `80` from any IP address on the server. That is the standard HTTP port, so Apache will use this block whenever a browser connects normally to the server.

`ServerName` and `ServerAlias` are the names Apache matches against the request's hostname. `ServerName your_domain` is the main domain for the site, and `ServerAlias www.your_domain` lets the same site respond to the `www` version too. This is what makes one configuration cover both addresses.

`DocumentRoot /var/www/your_domain` tells Apache where the website files live. When a request matches this Virtual Host, Apache looks in that folder for the page to serve, usually starting with `index.html` .

The `ErrorLog` and `CustomLog` lines send this site's errors and access records to log files. That matters because it keeps troubleshooting and traffic tracking tied to this specific website instead of mixing everything into one generic log.

One important gotcha: the names in `ServerName` and `ServerAlias` must match the domain you actually use in the browser or DNS, otherwise Apache may fall back to the default site instead of this one.

WAS THIS HELPFUL?

AI TUTOR ACTIONS

Ask a follow-up Explain more simply Show another example

Save and exit.

5.4 Enabling the Virtual Host

Enable the new virtual host file:

```
sudo a2ensite your_domain.conf
```

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Disable the default site:

```
sudo a2dissite 000-default.conf
```

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Check for configuration errors:

```
sudo apache2ctl configtest
```

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If everything is correct, you should see:

```
Syntax OK
```

Restart Apache for changes to take effect:

```
sudo systemctl restart apache2
```

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Apache should now be serving **your custom domain**.

6. Submission Guidelines

1. Take **screenshots** of the key steps:
   - Installing Apache.
   - Verifying the installation.
   - Configuring and testing the virtual host.
2. Compile screenshots into a **Google Doc or PDF**.
3. Upload the document to **Google Drive** and share the link with your instructor.
4. Ensure the instructor has the necessary **viewing permissions**.

🎉 **Congratulations!** You have successfully installed and configured **Apache on an Ubuntu Linux machine**! 🚀

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